THAMESMEAD REPORT

Alexander Pike

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The Project

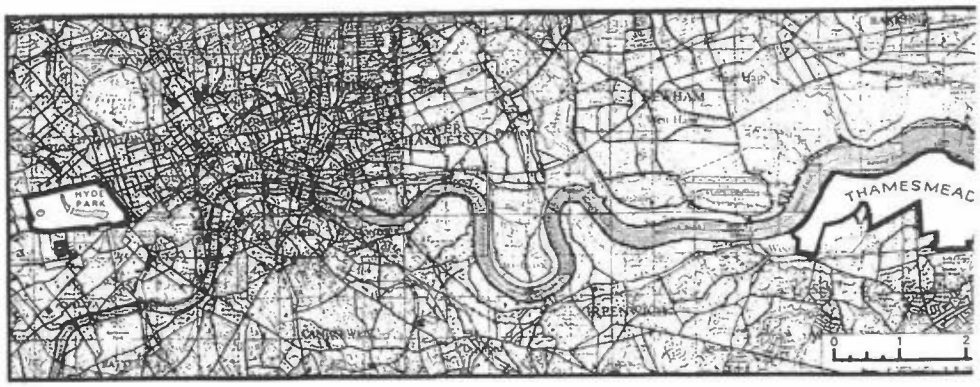
Initially started in 1962 as a project for 25,000 people on a 300 acre site in Bexley and subsequently enlarged by the addition of the Woolwich Arsenal site, the Thamesmead project now has 1300 acres available for development, of which 600 acres are zoned for residential use in the Draft Master Plan proposals. Aiming at a net density of 100 persons per acre and subject to reductions dependent on the ultimate balance between public and private housing, this will give a total maximum population of 60,000 people.

The site is a flat and featureless stretch of land on the south bank of the Thames with a three mile frontage to the river, but a number of fine trees and moated earthworks provide great opportunities for imaginative development. The scale of the landscape is large; facing the site, across a broad stretch of river, are Dagenham Docks and the Ford Works; within the site the proposed Ringway Two motorway is planned to pass through the area and under the river in a tunnel. This motorway with its intersection with the main spine road of the development has been a great formative influence in the design. It requires substantial areas of land, generates noise, and has a scale comparable with the natural landscape.

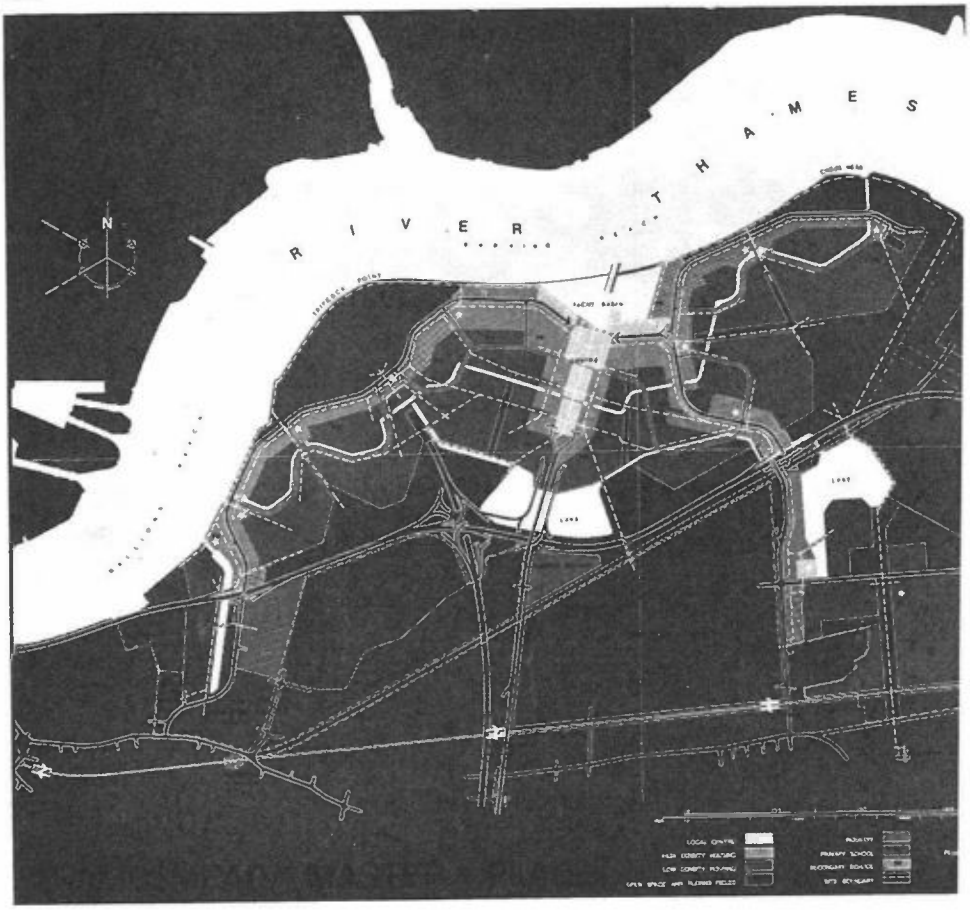
The decision to design Thamesmead to match this scale is the basic philosophy behind the scheme, into which all other requirements have been fitted. Thus the use of water has been conceived on a large scale, and linked with an extensive open space system; together they will provide facilities which, apart from local needs, will be of regional significance.

The other principles on which the scheme is based are:

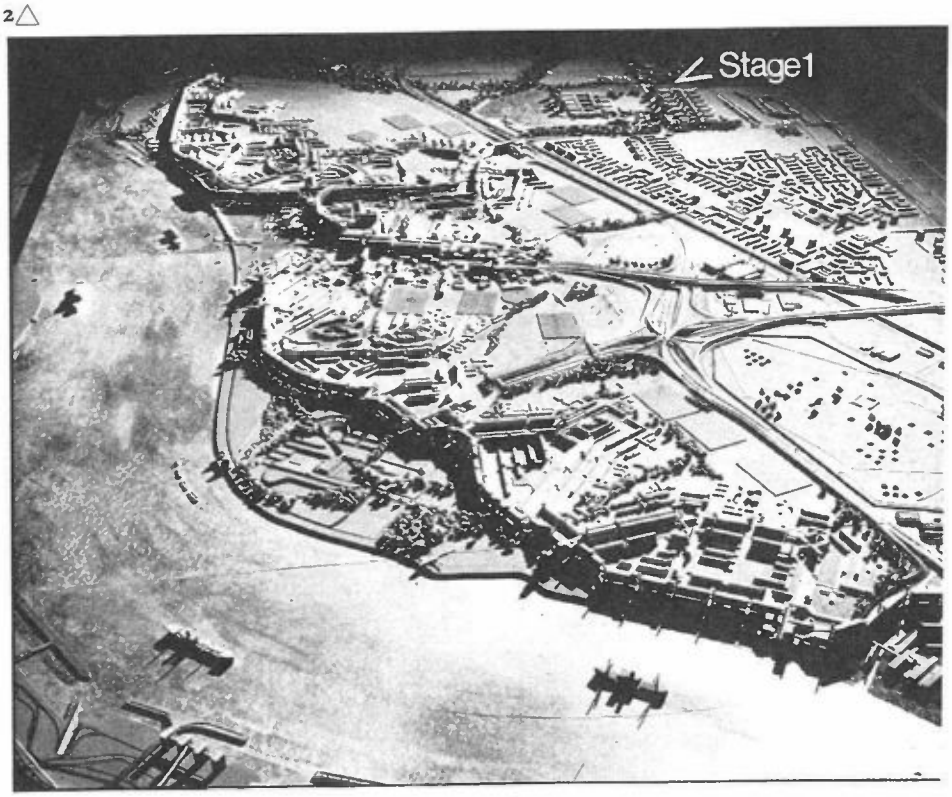
Thamesmead will not be a dormitory area. It is intended to provide a complete



1△



2△



1 *Location of Thamesmead in relation to central London and comparison in size with Hyde Park and Kensington Gardens.*
2 *Thamesmead master-plan showing the high and low density housing areas, industrial areas, railway and road systems, and the proposed tunnel site.*
3 *General view of model from west showing multi-level interchange and tunnel approach road in centre.*
4 *The first traffic plan (stages One, Two, and Three are tinted).*
5 *Stage One under construction, showing point blocks, linear housing, and 3- storey terrace housing.*
6 *Model of 5-storey linear housing, seen from the north, looking toward the sub-centre. (Model shows three blocks at sub-centre, built version has two).*
7 *Plan of Stage One.*

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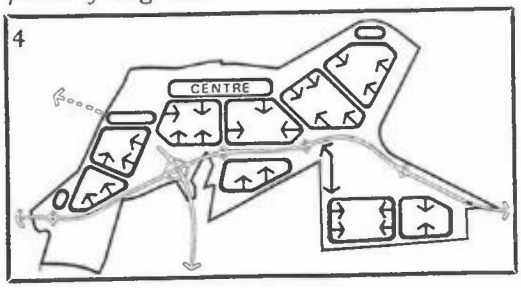
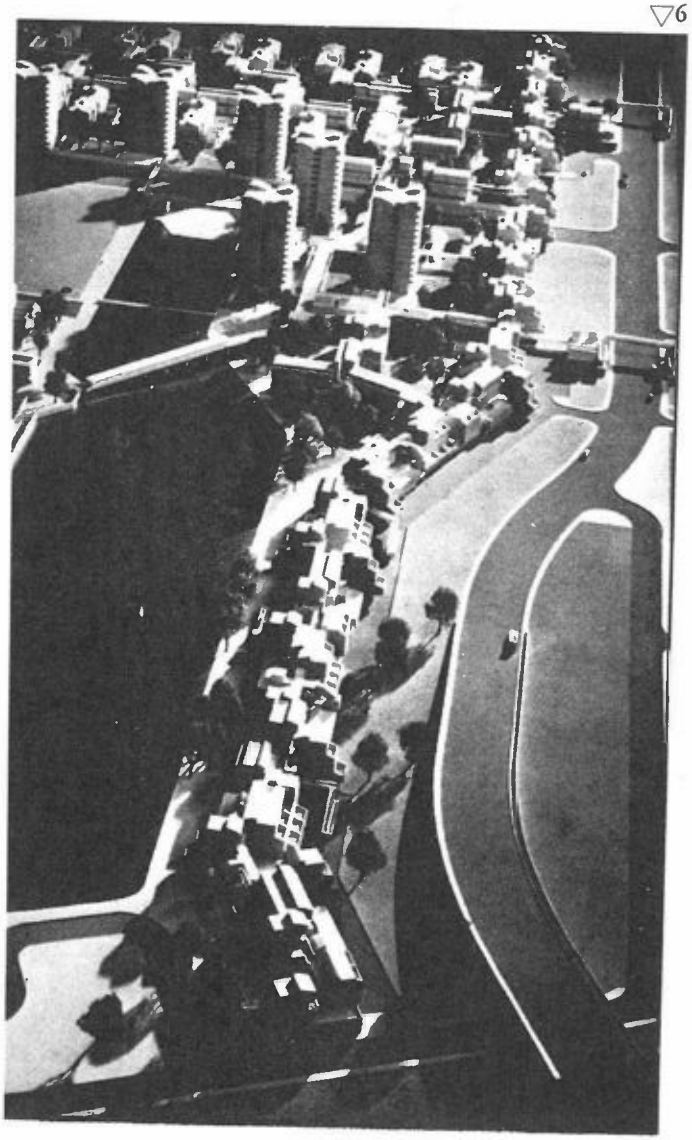
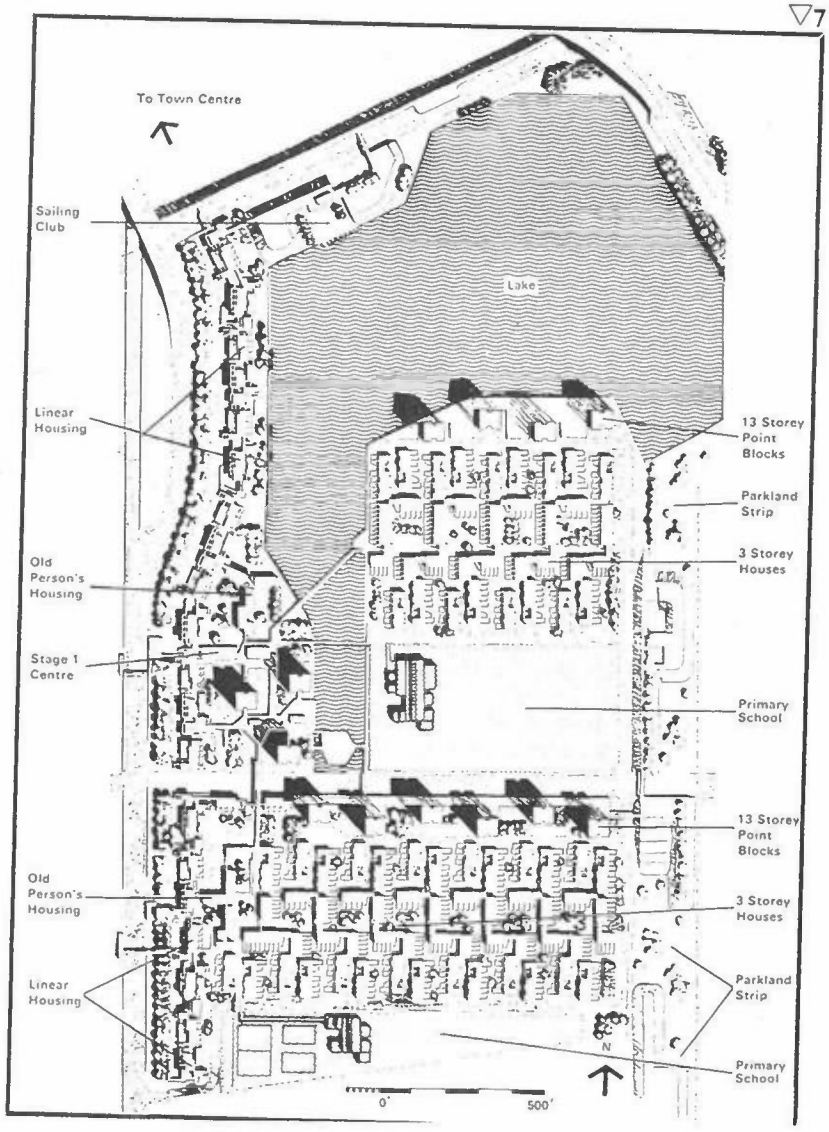


Photo: GLC

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▽7


To Town Centre
Sailing Club
Lake
Linear Housing
13 Storey Point Blocks
Parkland Strip
3 Storey Houses
Old Person's Housing
Stage 1 Centre
Primary School
13 Storey Point Blocks
Old Person's Housing
3 Storey Houses
Linear Housing
Parkland Strip
Primary School
N
0'
500'

environment with employment and full educational, communal, commercial and recreation facilities.

Within the relatively high average net density as many families as possible will have gardens or an outdoor living space. Density will be used as a design tool and will vary from housing areas with large family dwellings at 70 persons per acre to parts of the town where the density will be at about 140 persons per acre and provide dwellings for the small family and single persons.

Movement about Thamesmead for both pedestrians and vehicles will be along routes carefully designed to be safe and convenient. Communications between the new development and the neighbouring areas of London have been planned to ensure that the development will become integrated as part of Greater London.

It has been accepted that landscaping makes a major contribution to the quality of urban environment and the design of buildings and open space have been considered as a single problem.

The resulting design has a pronounced and coherent form with a line of tall buildings which in general run along the river frontage and a further high density leg of development running north–south from just north of Abbey Wood Station to the central area.

The centre is planned at right angles to the river, with a yacht basin at the northern end and a lake at the southern end, and is linked with the high density limbs of the housing areas.

The movement structure of the plan consists of a road network for vehicles and a footpath network for pedestrians (and possibly also for cyclists). These are planned to operate independently and are complementary to each other. Provision is also made in the plan for the introduction of public transport. The road network contains the main east–west spine road which forms a major feeder route to the Ringway Two motorway. There will be a system of main distributor roads running from this spine road along the high density housing areas and these roads will in turn serve a system of secondary distributor roads onto which will feed culs-de-sac giving access to the dwellings themselves.

The routes for pedestrians are being planned as carefully as the routes for vehicles. They will run independently of the roads, through open space, over and under roads, and also at an upper level through the high density development to link to the top of the river bank and the main level of the centre. These footpaths will serve schools, shops and playing fields so that movement on foot throughout the entire project will be pleasant and safe for all users.

The scheme will provide about 17,000 dwellings grouped into neighbourhoods which vary in size from about 1500 to 2000 dwellings. In general these neighbourhoods are designed with tall buildings on their northern or eastern edges and lower buildings within. They will contain their own primary school, a local centre for day-to-day shopping, churches, pubs, and some employment for local workers. The specially designed tall buildings will protect the central part of the housing areas from traffic noise and

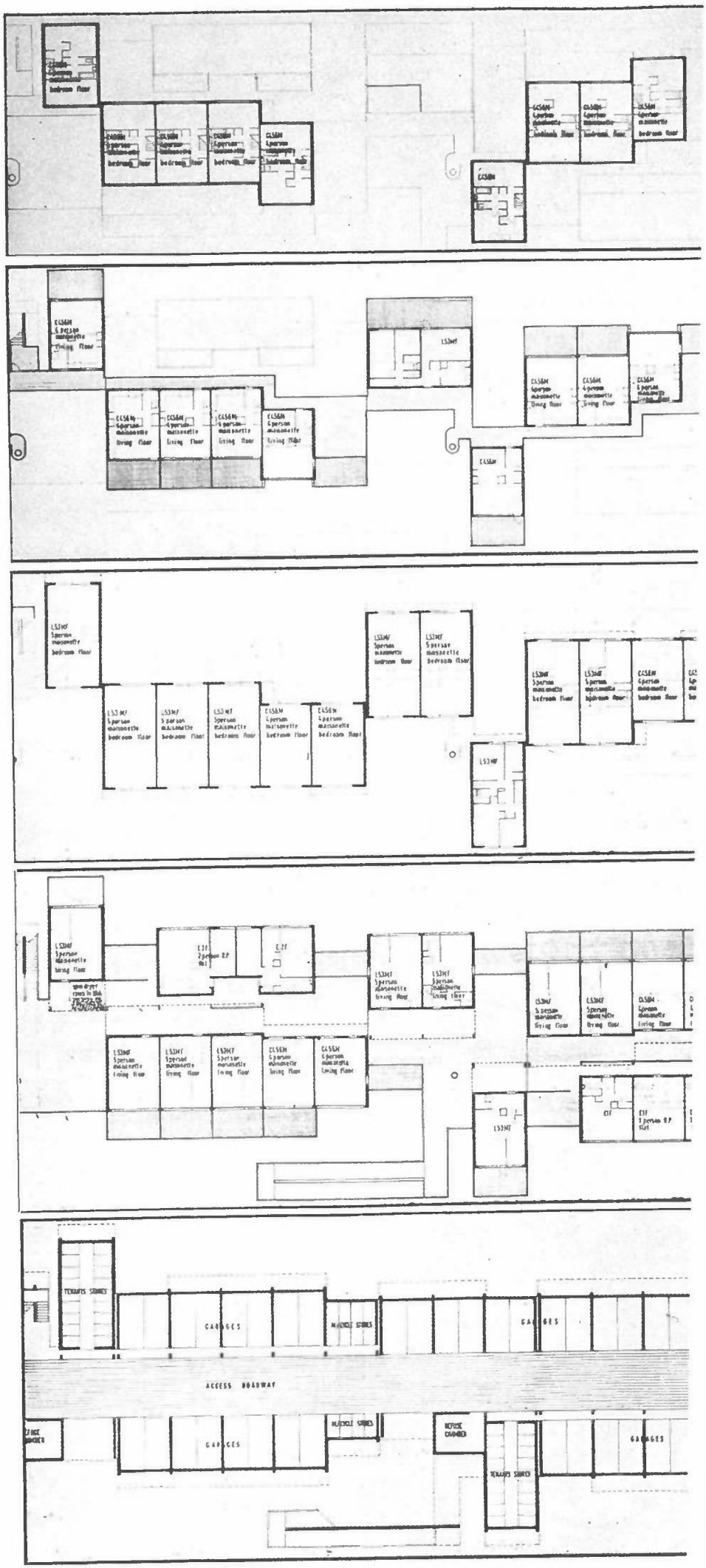
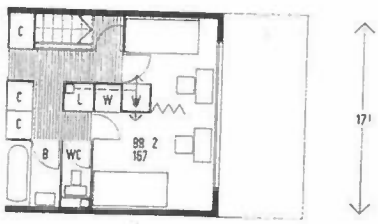
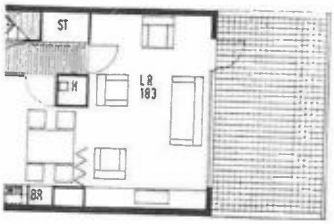
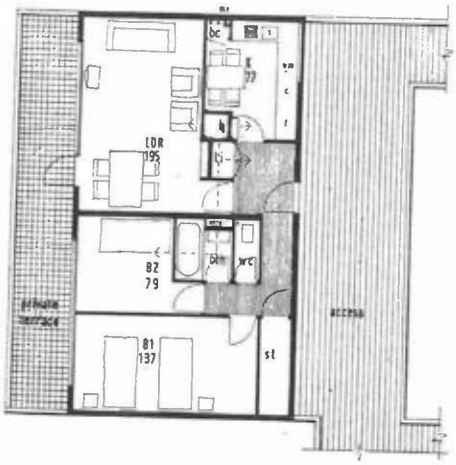


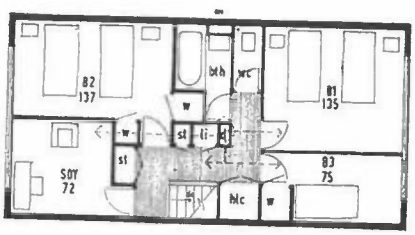


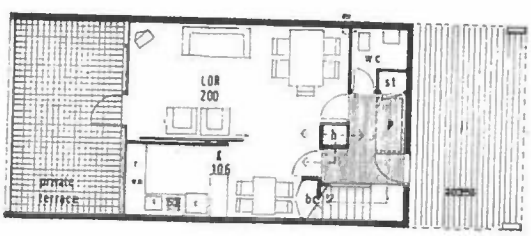
Photo: John Donat

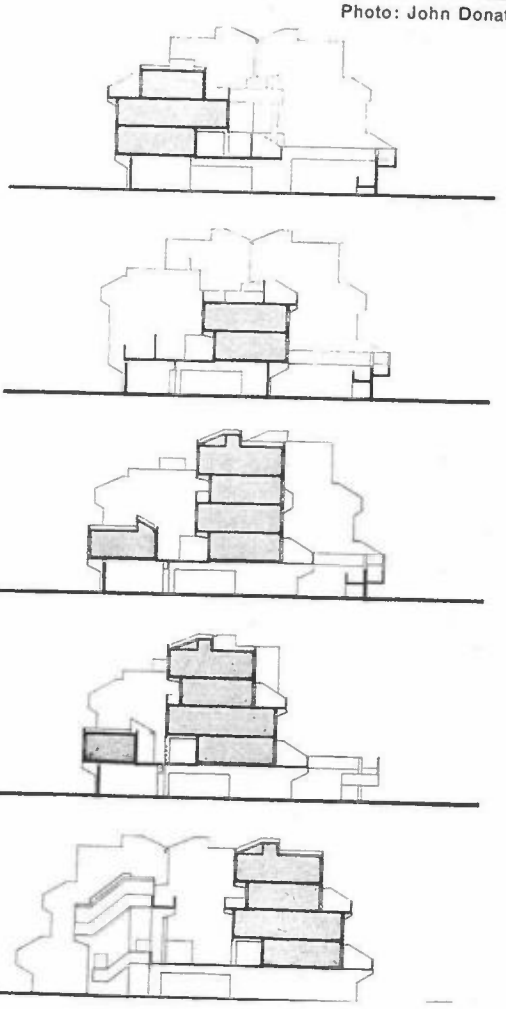












op *View of a completed section of linear*
e 4-person maisonette plan. Above right
lan. Below right *5-person maisonette plan.*
sections through linear housing.
near housing floor plans.

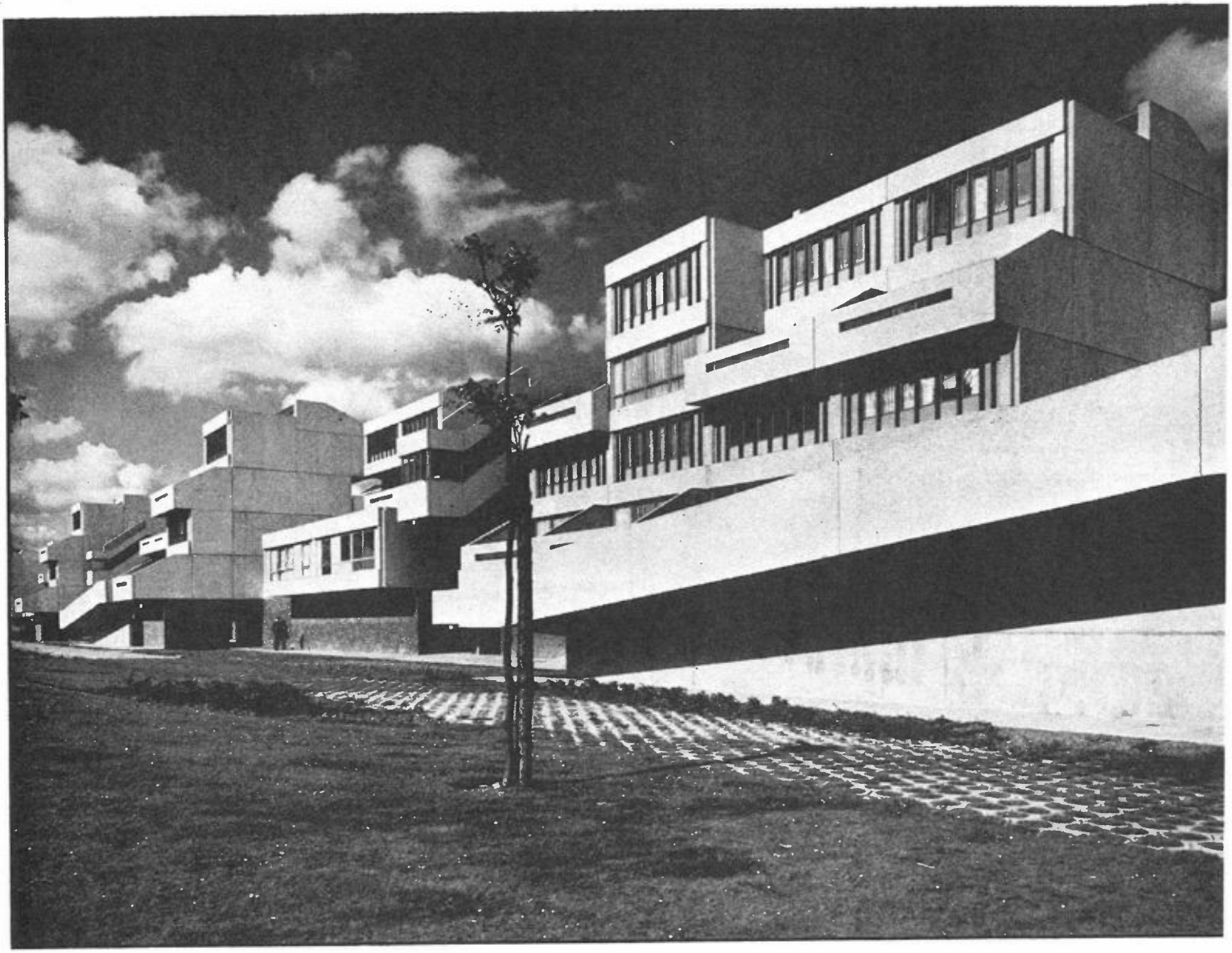
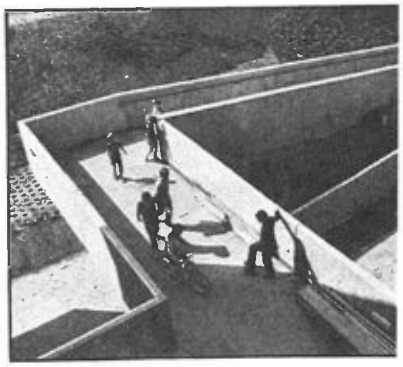
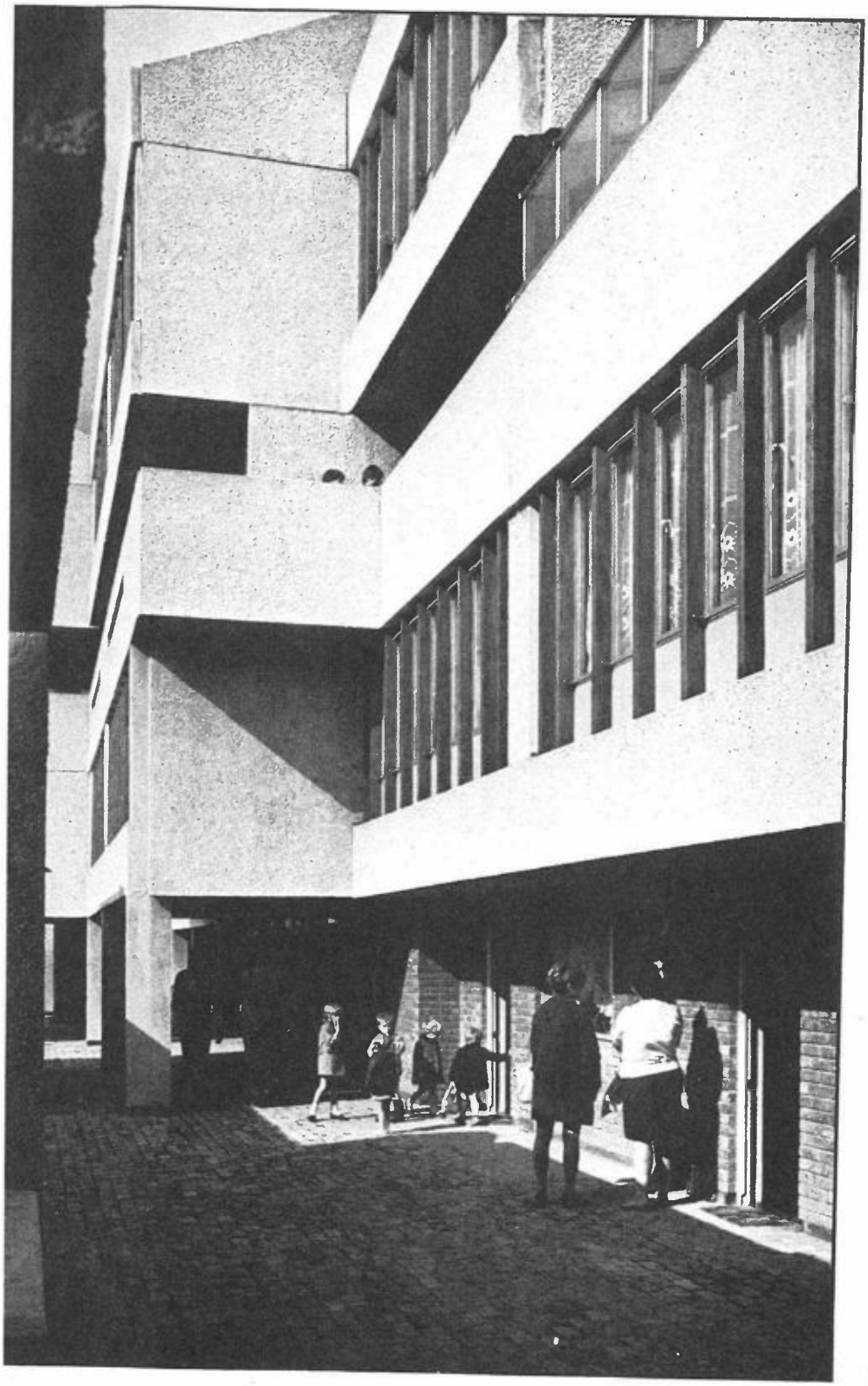


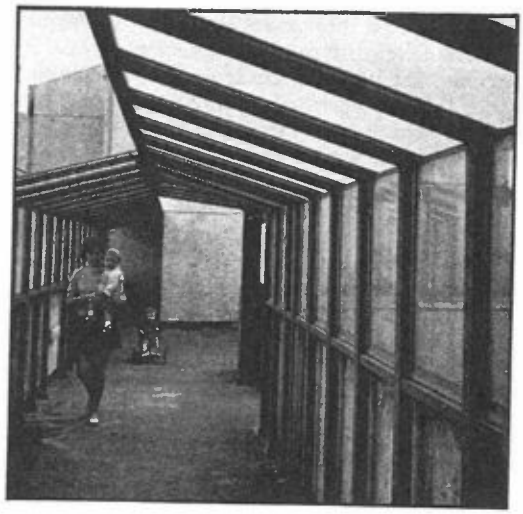
Photo: GLC

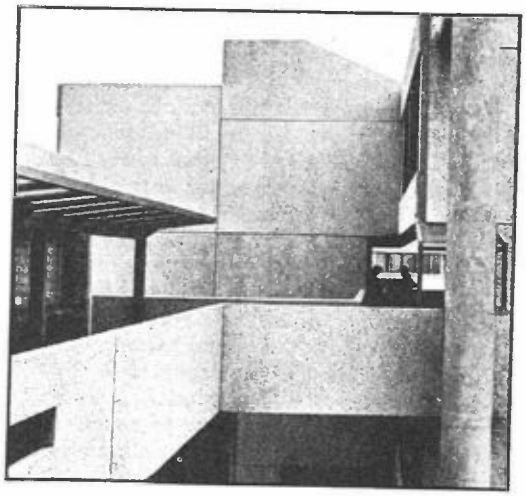


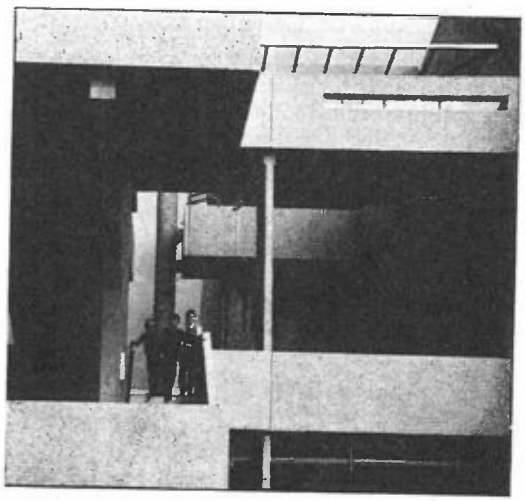


















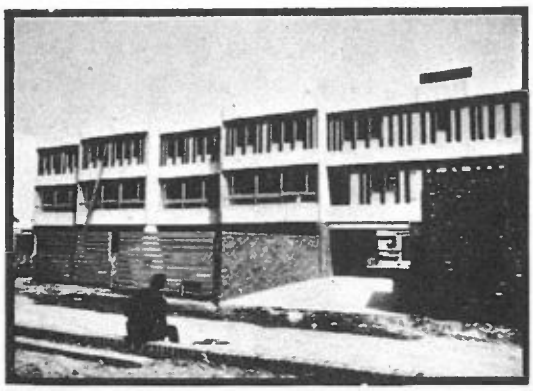
wind. Beyond these areas the major school playing fields and public open spaces will serve to separate the lower density housing from the main road of the development.

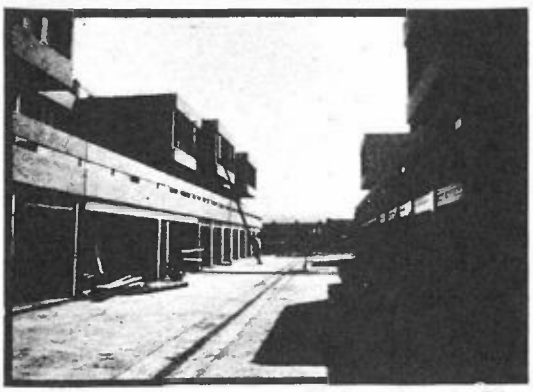
The plan makes extensive use of water throughout the site. Two lakes will be created, linked by a canal forming part of an entire system running through the town. These canals will run alongside the secondary road serving the housing and will act as a barrier to crossing the road at unsuitable positions. Thus it will be possible for pedestrians to move across the roads and canals only at selected points where footpaths go over the canals and under the roads, which will be slightly raised at these points. The excavated material from the lakes and canals will be used to create differences of level within the open space system.

The lakes will form part of the general open spaces in the project and these, associated with extensive tree planting, are already well in hand.

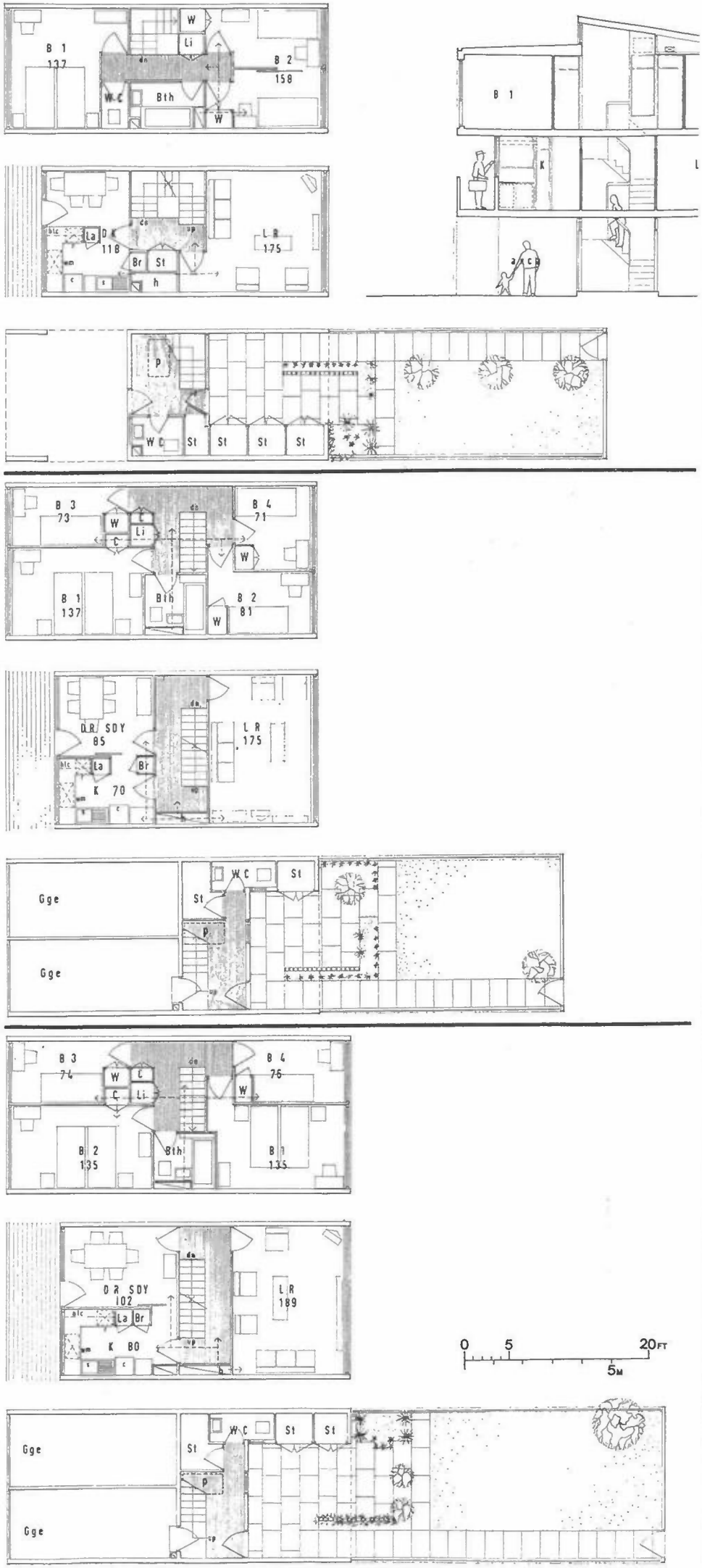
It is anticipated that the main element of the recreational provision will be the yacht basin. This is proposed to be of international status and aims to bring to the centre of Thamesmead, and to this part of London in general, a facility and colourful activity to add greatly to the attractions of the area. The basin will be flanked by high density housing and central area uses.

The central area will contain shopping, office employment, community, recreational and educational facilities including a comprehensive medical centre and an ecumenical centre. The design of this area provides for a number of dwellings and a high level main concourse with vehicular services, parking and a public transport interchange at natural ground level. The main concourse will provide access to, and views of the river at the northern end and the lakes at the southern end. It is suggested that important educational buildings are placed at this end of the centre, to overlook the lake and also to form an interesting approach to the centre when using the main road over the lake.





Photos: Alexander Pike





wo photographs of the recently completed nd a typical floor plan.

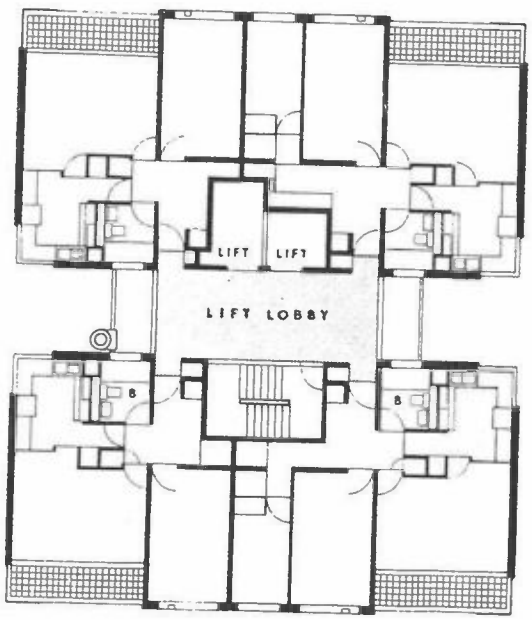
Building system chosen

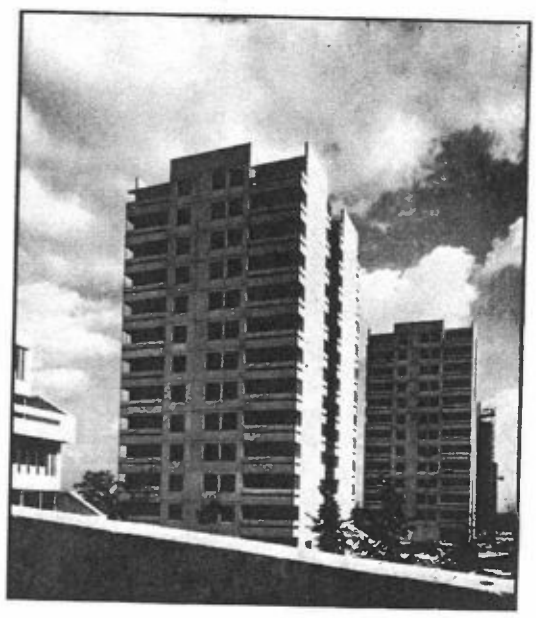
The Master Plan was prepared in 1965/66 followed by a Second Master Plan later in 1966. Considering the exceptional scale of the project, one of the fundamental objectives was to start building as quickly as possible. It was decided that to achieve this aim an Industrialised Building system should be employed, but that this should not be regarded as a restriction on the design. Previous experience had shown that the use of Industrialised Building systems had produced overall costs that were slightly above those for traditional construction and revealed little or no indication of the profit margins.

The design team decided that the correct approach would be to design a good project without reference to a specific IB system and then apply the use of a system to the scheme.

By the Spring of 1966 design work on Stage 1 was sufficiently advanced for a cost plan to be established by the Council's quantity surveyors. This was intended to cover the first two stages and was based on normal traditional construction. Advice was then sought from the National Building Agency for their opinions on which combinations of contractors and IB systems could show suitable production and management capacity for the project. After initial discussions with a number of possible combinations, three national building firms were invited to prepare submissions to the Council, showing how they would propose carrying out the work with their particular system within the cost targets.

The contractor finally chosen was Cubitts, whose submission was based on the use of the Balency & Schuhl system, which the company had already used on the extensive Ballymun housing project in Dublin.





op 4-person house plans and cross-section, house plans, and below *6-person house plans. iph shows completed section of linear housing, otograph shows rear, with elevated walkways*

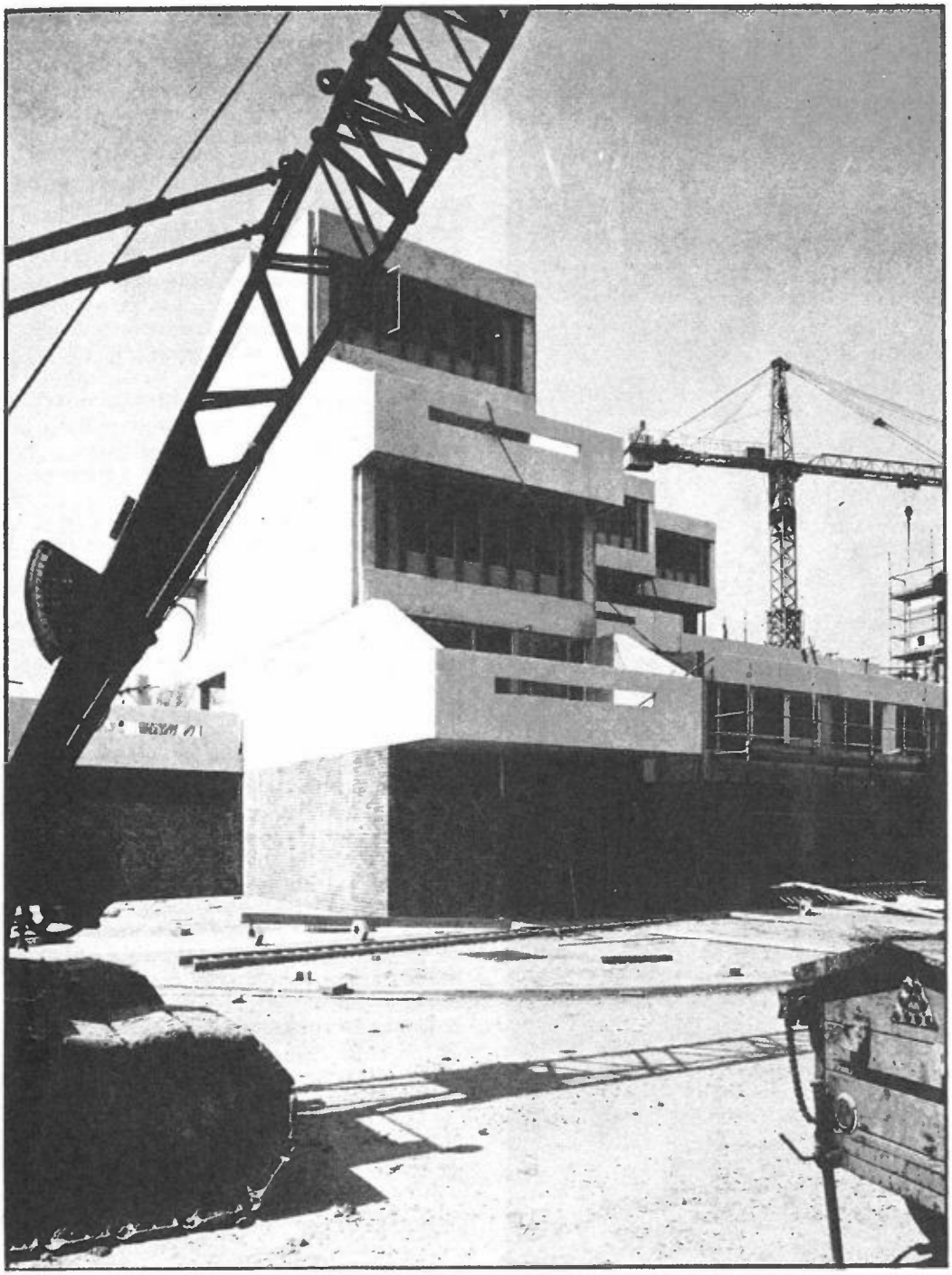


Photo: GLC

The Balency system

The Balency system has been used on the Continent for over 15 years and is claimed to be one of Europe's best-tried IB methods. It is based on the use of precast concrete wall panels, but differs from almost all other 'heavy' concrete systems in that it normally employs an *in situ* concrete floor slab construction. Both the internal and external wall panels are designed as load-bearing elements, precast in highly sophisticated steel moulds. The height of the wall panels is constant for any one contract, but considerable latitude is possible in the length of panels.

The basic principle of the system presupposes that each building will have a total dead weight sufficient at all times, whether during construction or completed, to resist the lateral wind load forces without inducing undue tension in any of its constituent parts. Each precast wall is designed as a load-bearing unit to carry its own weight together with all the super-imposed weights and a proportion of the horizontal wind load.

All internal walls are cast vertically and have electrical conduits and outlet boxes cast in. Special wall units, known as 'technical blocks', have pipework for hot and cold water services and drainage cast in, with kitchen outlet points on one side and bathroom outlets on the other, a technique claimed to reduce the site work required for the installation of piped services.

External wall panels are cast horizontally and consist of a 3in outer skin of concrete with exposed aggregate finish, a 1in thick expanded polystyrene insulation membrane and a 5in thick internal structural concrete skin. The two skins are tied together by metal ties which pass through the insulation membrane.

The standard method of floor construction is for all items to be cast in, conduit runs, outlet boxes, reinforcement, etc., to be assembled on the ground on a metal frame made up of scaffold tubes. The unit is then lifted up into position ready for the concrete to be poured.

This method was employed throughout stages of the Thamesmead project but modified so that the entire floor slab c be precast and positioned as one unit three-storey housing.

When all walls for one storey have erected and supported, formwork is p in position and the floor slabs cast ov walls. The tops of walls are arranged project into the floor slabs ½in above level and all protruding reinforcement tops of wall panels is bent down for c into the floor slabs, ensuring rigid joi

The vertical joints between external panels form hexagonal holes which ar carefully hand-filled with dry-mix con A loose tongue of felt-covered expanc polystyrene is fitted within the joint t maintain insulation and to prevent co seepage into the front of the joint.

The contract

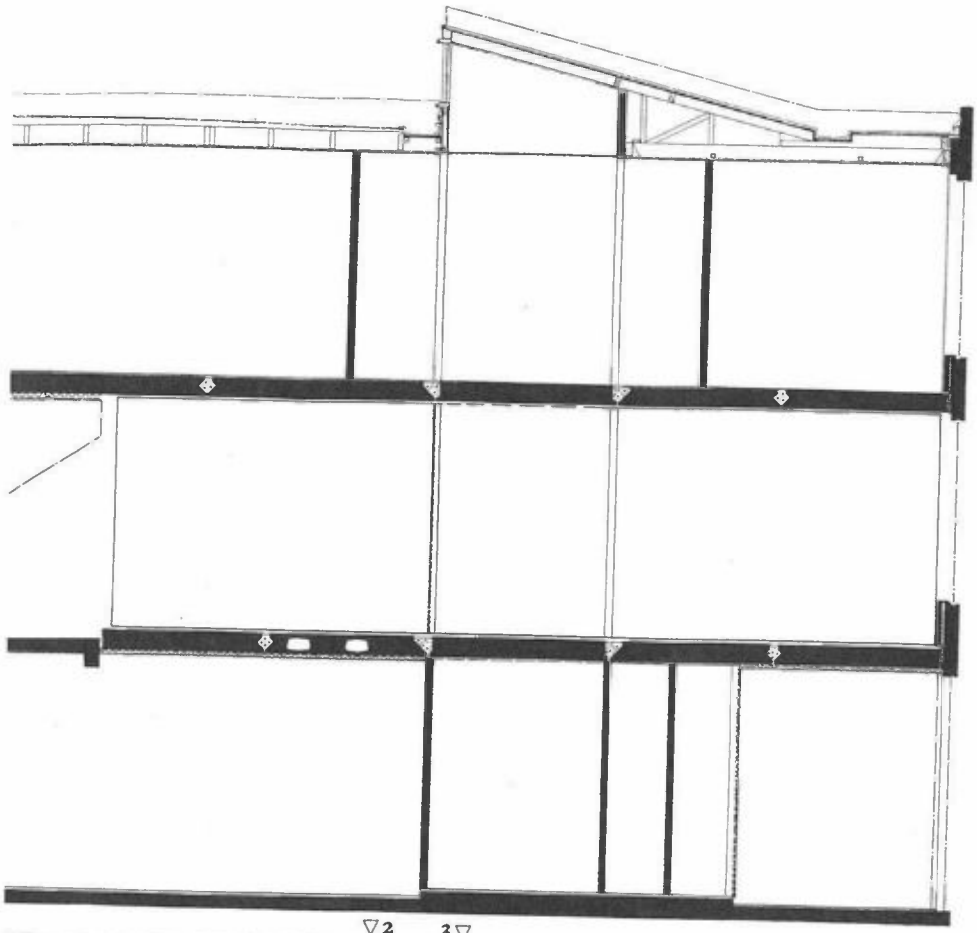
The exceptional size of the project de special attention to the form of contr and in this instance the unusual Valu system seemed to be particularly eligi spite of the fact that it had not hither employed for an IB project.

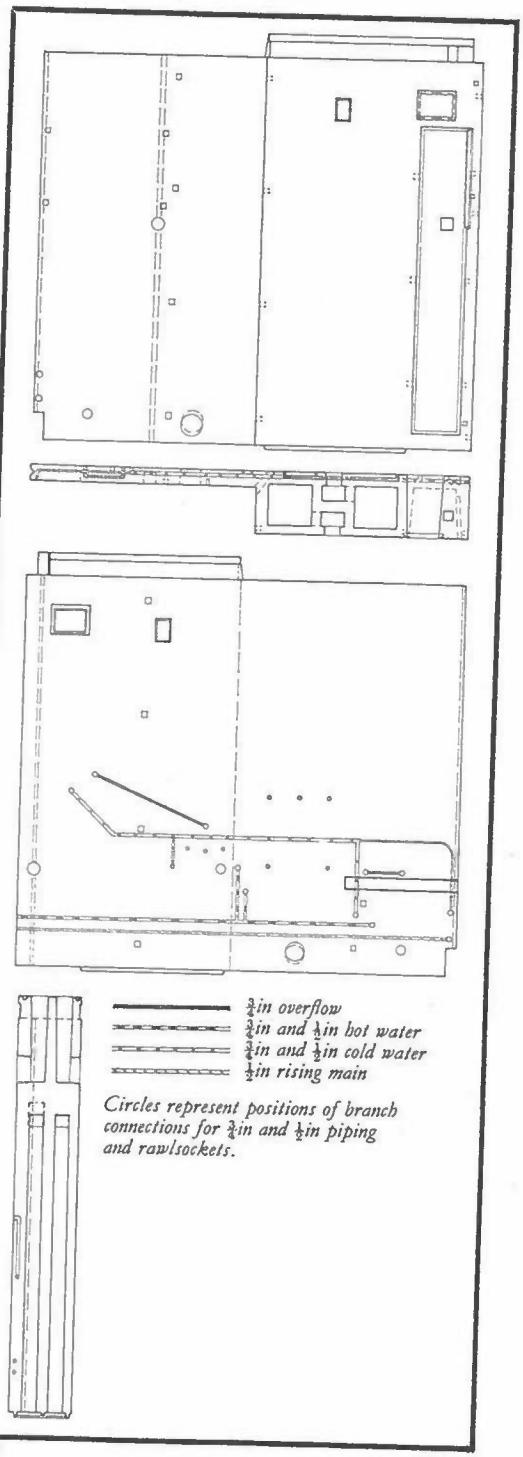
This system was devised in 1920 to the need for large housing contracts time when prices of labour and mate were constantly fluctuating and condi the building industry were unsettled. essence, the contract provides for the building construction to be financed Council, the contractor being paid a cover his head office expenses. Tende be invited from selected firms of suit experience and capacity who quote p centage variations to a fixed schedule rates and prices or, alternatively, sch of items of work with indicative qua may be provided for the tenderers to their own prices and rates. Sometime offer is negotiated with a single cont to carry out the work on an agreed s

The actual cost of the work is com with a 'value' calculated from the sch and agreed with the contractor. If 'v and 'cost' correspond, the contractor fee is computed as a percentage (abo per cent) of the 'value'. If the actual is more than the 'value' the fee is rec but if less than the 'value' the contra paid a bonus, representing part of th saving, in addition to his basic fee. T comparisons between cost and value made at intervals throughout the pro of the work and at the completion o contract.

The economic advantages of this t of contract increase with the size of scheme, and costs compare favourab other types of contract for large scal schemes. It also enables work to be more quickly, it being unnecessary f detailed layouts, working drawings of quantities to be available before Variations and modifications can be porated and different types of buildi be introduced.

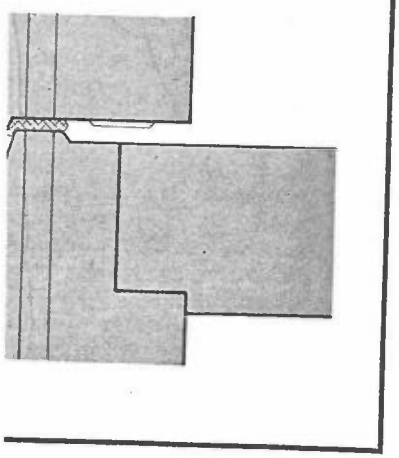
As the Council had previously use type of contract solely for schemes e ing traditional building methods, a s sidiary reason for its use on this pro was to obtain important data on the mic aspects of Industrialised Buildin subject on which comparatively littl information is available.

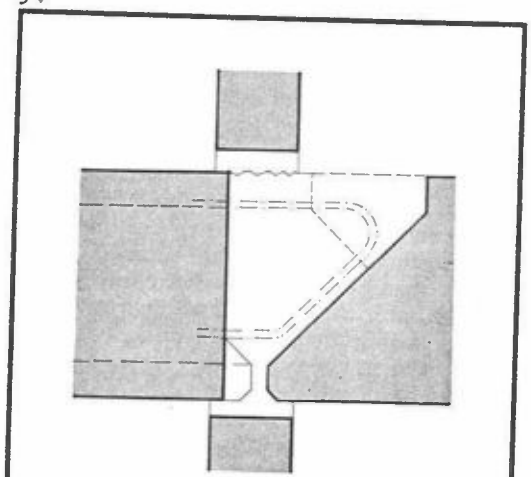


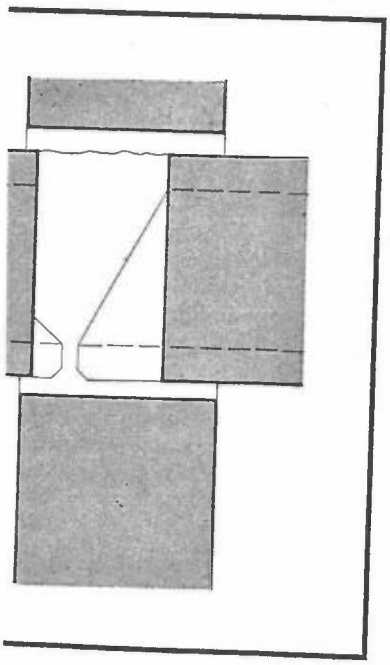


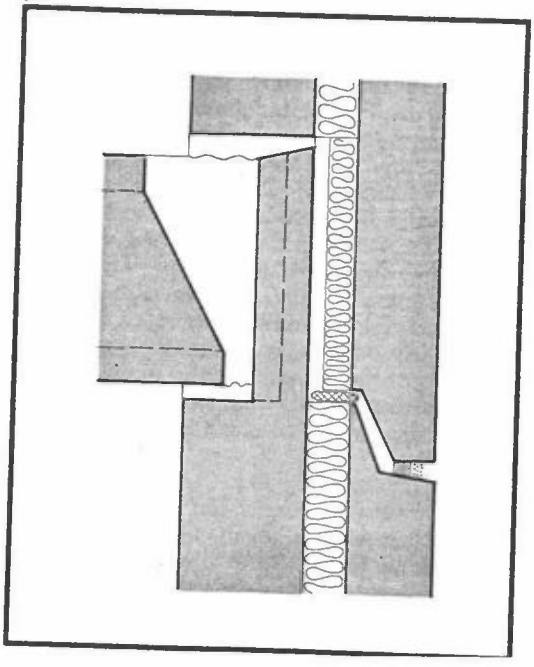
¾in overflow
¾in and ½in hot water
¾in and ½in cold water
½in rising main

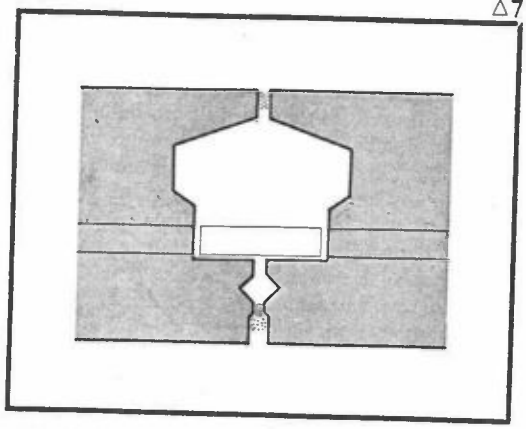
Circles represent positions of branch connections for ¾in and ½in piping and rawlsockets.











1 *Section through 5-person 3-storey house.*
2 *Horizontal joint between external wall panels.*
3 *Connection between floor slabs.*
4 *Junction of floor slabs at cross wall positions.*
5 *Junction for floor slab and external wall.*
6 *Vertical joint between external wall panels.*
7 *Special wall panel, termed 'technical block', incorporating all service and drainage pipework. Shown is the point block unit.* (Top *kitchen elevation*, lower *bathroom elevation*) *with horizontal and vertical sections through fl...*

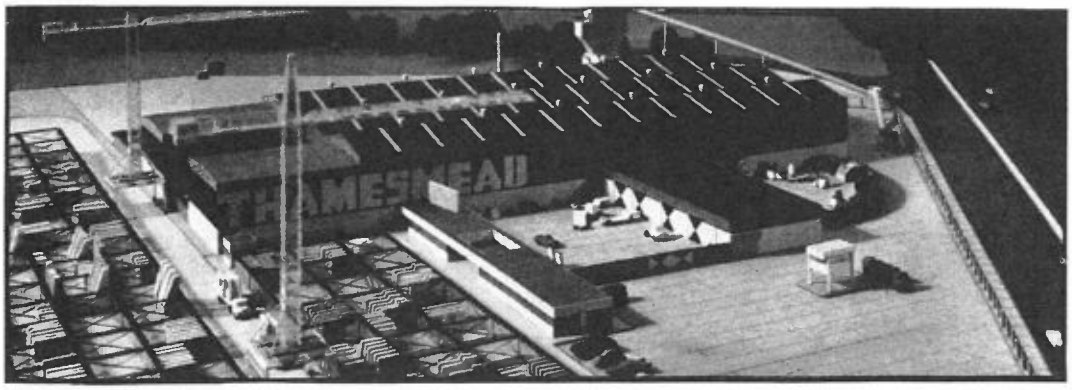


Photo: GLC

The first phase

The contract for Stages I and II of the project was awarded to Cubitts in October 1966, for the construction of 4000 dwellings over a period of five years at a cost of nearly £30 million.

Stage I has an area of 85 acres, of which 50.7 acres are to be used for housing and the remainder allocated for two primary schools, the sub-centre provisions, other minor communal buildings and a lake of approximately 30 acres. This stage will provide approximately 1500 dwellings in three different housing types; 13 storey point blocks; 2 to 5 storey linear maisonette blocks; 3 storey terrace houses. The twelve point blocks are sited in two groups; four on the south bank of the main portion of the lake and eight in the middle of the site. Six of these are on the southern side of the main east–west roadway and the remaining two on the opposite side of the road, directly related to the Stage I sub-centre.

The linear housing runs down the full extent of the western side of the site, following the line of Harrow Manor Way, with extensions of the pedestrian deck to the sub-centre and the line of six point blocks at first floor level. The 3 storey terrace houses are grouped to create pedestrian enclosures within the perimeter access road pattern and are sited on each side of the central primary school site.

The 5 storey linear blocks are of a type not previously used, and have been specially designed for this project. A ground level access road gives access to the garages, above which the linking pedestrian level route, half a mile through the first stage, gives access to the four floors of maisonettes and the one-storey old persons flats located at deck level.

Preliminary design studies at the tender stage indicated that the highly modelled form of these blocks made them unsuitable for the economic application of the Balency system, and the first of these units on the site have been built by rationalised *in situ* crosswall and floor slab techniques, with non-loadbearing cladding panels. This decision was reinforced by the added advantage that an earlier start could be made on this part of the work without waiting for the completion of the factory for the production of the precast components

Now that the factory is in production, future elements of the linear blocks will employ the Balency system units.

Site conditions

The site is low-lying with a high water table and lies below the high-tide level of the Thames. The sub-soil is of poor load-bearing capacity with an under-laying stratum of peat of varying thickness. Pile foundations will be necessary for all buildings, down in some cases to depths of 30 to 40 feet. The former by-law restricting habitable floors to a minimum of 8ft 6in above Newlyn Datum has been superseded by the new Building Regulations, but may be retained in this instance as a planning regulation. The final decision is dependent on the progress of proposals for the Thames River Defences and the possibility of a Thames Barrage.

The site allocated for Stages I and II of the development is on the Erith Marshes, which only drains at low tide and the lake planned for this area will facilitate drainage and act as a flood water balancing area.

The site factory

The on-site factory (photo above) for producing the concrete units was the first building to be erected on the site. It has an area of 114,000 square feet, a production line 560 feet long, and using one-shift working has an output of approximately 950 dwellings a year. This could be increased to 1500 dwellings a year by using two shifts.

The units are produced in 50 steel moulds (12 vertical, 38 horizontal and special moulds) which can produce approximately 700 different castings. The steel end formers, which make up the side walls of the mould, can be fixed at any predetermined position of bolt holes in the baseplate of the mould, thus allowing the production of a limited number of different sized units from the same mould.

At the present rate of production each horizontal mould produces on average two castings a day and each vertical mould three castings a day. This is achieved by accelerating the initial curing period of the concrete by heating both the concrete mix (to about 120°F) and the moulds (to about 140°F) before pouring. Each horizontal mould is fitted with an hydraulically operated autoclave head which when lowered completely covers the baseplate and the concrete panel. The average initial curing period varies from five hours for the floor slabs and four hours for the external wall panels in the horizontal moulds to three hours in the vertical moulds. These periods vary according to the temperature in the mould, the temperature of the mix when poured, the ambient temperature and the size of the panel.

The special moulds include those designed to produce stairs, balcony fronts, floor slabs (for the three-storey house only) and technical blocks.

Vertical moulds are used to produce internal partition panels. The internal faces of these moulds are constructed of polished steel to achieve a smooth panel surfa direct application of paint or wallpap after erection. The concrete is injecte pneumatically from the bottom of th vertical moulds to give greater contr over the placing, to avoid distortion steel and segregation of the mix, and allow for natural displacement of air the mould.

External wall panels are cast flat in zontal moulds. The panel is placed w the external facing mix laid on a reta applied to the baseplate, over which the concrete, reinforcement and insul The top (internal) surface of the unit power-floated twice during the initia period (one hour and two and a half after placing the mix in the mould) t a finish comparable to that obtained partition panels cast in the vertical m On completion of casting, the autocl hood is raised, and all end formers, e that along the bottom edge of the pa are removed and the baseplate is tilte hydraulic jacks to allow the overhead crane to remove the unit in a vertica position to the finishing bay.

In the finishing bay the aggregate exposed by hosing and brushing off retarder coated surface; all units are and any necessary minor repairs are out.

The finished units are lifted from finishing bay by tower crane and pla the stockyard for a minimum period hours before being loaded on to trar for delivery to the building site.

Costs

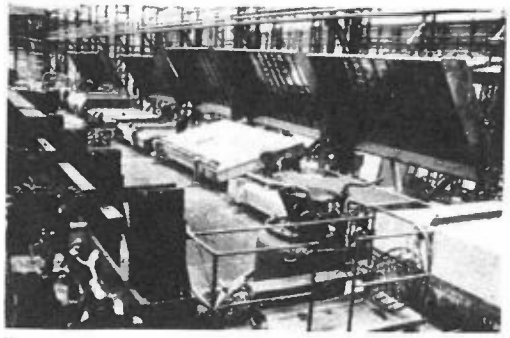
The immense scope of the project ar type of contractual system employed assessment of the overall cost extrem difficult. The initial cost plan was fo before the introduction of the preser Ministry housing yardsticks, and fut phases of the work will require mod to conform, in spite of the objective obtain conclusive data on the cost o Industrialised Building.

The cost of the site factory for ma facturing the components will be wr during the production of the 4000 d comprising the first phase. Thereafte costs will be related merely to the g maintenance and replacement of plar equipment, and production cost for components will fall sharply, consid reducing the overall costs of dwellin

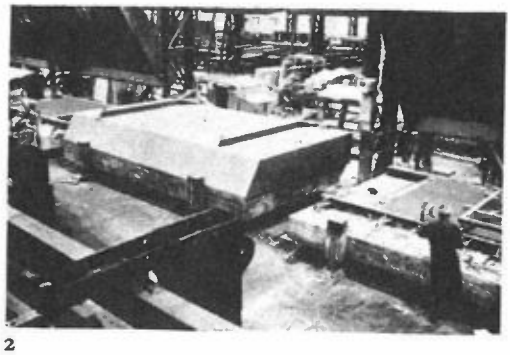
The extent of the scheme and the and basically unsuitable, nature of th have, predictably, produced high ab in the cost plan. Site development c high for a number of reasons:
the drainage system is complex and tensive; large areas of the site will r building-up;[1] piling is necessary eve not only for buildings, but also for elevated walkways and bridges and the sewers of the main drainage syst the cost of high capacity services su and long distribution runs is consid the lakes, lagoons and canals, necess this case for the drainage system, co an extra not normally encountered.

In addition to these items certain

[1]The entire 50 acre housing site for Stage 1 covered with sand for a depth of two feet be could commence. The site for Stage 2 was t same way, the sand being pumped from bar Thames.



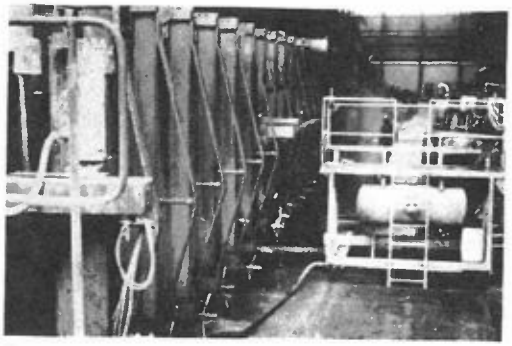
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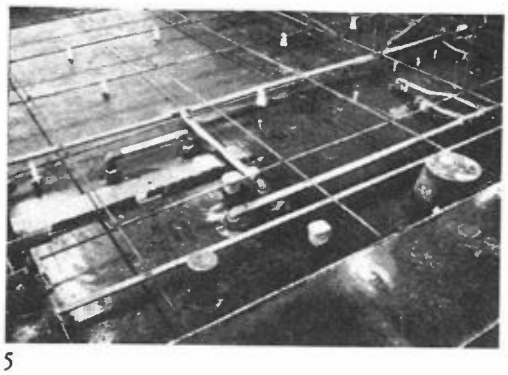
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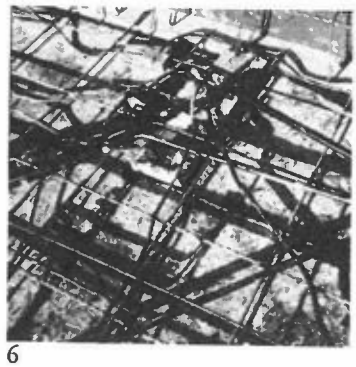
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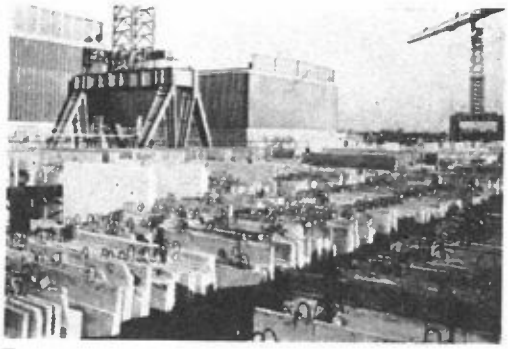
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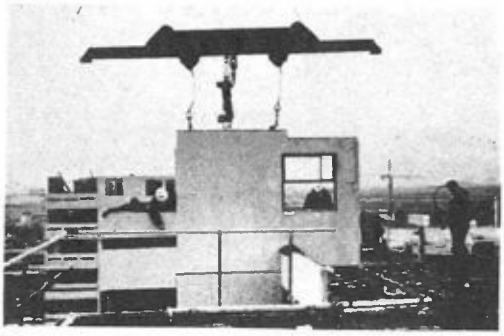
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8



9



10

The manufacture of the concrete units

1 *Interior of factory showing horizontal moulds for Balency panels.*

2 *Horizontal mould with autoclave hoods in raised and lowered positions.*

3 *External wall panels being removed by gantry crane after baseplate of horizontal mould has been tilted.*

4 *Row of vertical moulds for internal wall panels. The concrete is injected pneumatically from the bottom of the mould.*

5, 6 *The floor slab grillage incorporating all cast-in items and reinforcement is assembled on the ground on a metal frame mad tubes. For the earlier dwellings on this was lifted into position and t cast* in situ. *The current practice slab on the ground and hoist as a*

7 *Completed panels are transported yard for final curing.*

8 *In the finishing bay the panels are the retarder coated surface brushe the Norwegian granite aggregate.*

9, 10 *After curing and finishing th panels are taken from the stockyar and hoisted by tower crane and pl*

Photos: Alexander Pike.

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principles—the provision of extensive elevated walkways and the use of single aspect dwellings to contain the noise from major roads—are an inevitable source of increased costs. In mitigation of these abnormals, land costs are low at approximately £170 per dwelling compared with an average between £1000 and £1500 per dwelling for all other GLC housing sites.

The future

The interest of Thamesmead lies not merely in the exercise of urban planning on a vast scale. It seeks to provide solutions to the national problem of increasing the output of building within the limits of the total labour force available.

Of the £225 million estimated to be the total cost of the project, approximately £180 million will be building work, extended over a 13-year period. The present rate of building of £7·5 million a year is already very high, probably exceeding twice the rate ever previously achieved in this country, but even this must be improved if the target programme is to be met.

The estimated figure for the number of operatives on the site is 3500, but as it is difficult to attract certain types of tradesmen to the area, 2500 is probably a more realistic figure. The total of 17,000 housing units will be provided by the GLC and private developers in the ratio of about 65 per cent to 35 per cent, giving the GLC responsibility for about 11,000 units, an output well within the capacity of the site factory.

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